Technical Specification & Engineering Cut Sheet

Cradle-to-Grave Life Cycle Inventory for Environmental Product Declaration (ISO 14025 / EN 15804+A2) — Modules A1–A5, B1–B7, C1–C4 and D

Product Name: Carrier AquaEdge 19DV Centrifugal Chiller
Model Capacity: 500 RT (1,758 kW Nominal Chilled Water)
Refrigerant Type: R134a (Initial Factory Charge: 45.0 kg, fugitive leak rate: 2.0%/yr)
Plant Utility Power: 34,000 kWh annual manufacturing electricity allocated to unit
Total Dry Shipping Weight: 4,420 kg
Manufacturing Site: Assembly Plant, Chicago IL (Grid Region: US Average)
Installation Site: Commercial District Cooling Plant, Dallas-Fort Worth TX (Grid: US-ERCOT)
PCR Category Standard: UL 10010-4 Part B & EN 15804+A2 (Industrial HVAC Systems)
Declared Unit: 1 unit of 500 RT water-cooled chiller
Reference Service Life (RSL): 25 Years with standard maintenance intervals

Module A1 — Raw Material Supply (Bill of Materials)

Component	Material	Weight (kg)	Supplier	Transport (km)
Compressor Volute & Casting	Structural Steel	2,250	Midwest Iron Works	420
Evaporator & Condenser Tubes	Copper	720	Great Lakes Copper Corp	280
Semi-Hermetic Induction Motor	Electric Motor	480	ElectroMotors Industrial	650
Shell Thermal PUF Insulation	Polyurethane Foam	140	FoamTech Insulation	190
Power Inverter & Solid-State VFD	Power Electronics	110	Advantech Power Systems	890
Microchannel Subcooler Coil	Aluminium Alloy	95	Alloy Heat Exchangers	350
Refrigerant High-Pressure Piping	Stainless Steel 304	85	Precision Alloy Tubing	340
Base Skid Frame & Mounts	Structural Steel	540	ArcelorMittal Midwest	260

Total BOM mass: 4,420 kg. Cut-off: components below 1% of total mass excluded per UL 10010-4.

Module A2 — Inbound Transport to Manufacturing Plant

Leg	Route	Transport Mode	Distance (km)	Freight Mass (kg)
A2-01	Midwest Iron Works -> Chicago Plant	Heavy Lorry >32t (EURO 6)	420	2,250
A2-02	Great Lakes Copper -> Chicago Plant	Heavy Lorry >32t (EURO 6)	280	720
A2-03	ElectroMotors Industrial -> Chicago Plant	Freight Train (Electric/Diesel)	650	480
A2-04	FoamTech Insulation -> Chicago Plant	Heavy Lorry >32t (EURO 6)	190	140
A2-05	Advantech (Kaohsiung -> Long Beach)	Transoceanic Container Ship	10,800	110
A2-06	Advantech (Long Beach -> Chicago)	Freight Train (Electric/Diesel)	890	110
A2-07	Alloy Heat Exchangers -> Chicago Plant	Heavy Lorry >32t (EURO 6)	350	95
A2-08	Precision Alloy Tubing -> Chicago Plant	Heavy Lorry >32t (EURO 6)	340	85
A2-09	ArcelorMittal Midwest -> Chicago Plant	Heavy Lorry >32t (EURO 6)	260	540

Module A3 — Manufacturing & Assembly

Input / Output	Value	Unit	Notes
Grid Electricity (US Average)	34,000	kWh	Annual plant electricity allocated per unit
Natural Gas (Process Heat)	18,500	MJ	Paint curing oven & space heating
Process Water	52.0	m3	Hydrostatic pressure testing & wash-down
Wooden Shipping Skid (Packaging)	120	kg	Kiln-dried softwood timber
LDPE Stretch Wrap (Packaging)	15	kg	Weather protection film
Steel Machining Scrap	185	kg	Sent to recycling (EAF scrap)
R134a Factory Charging Loss	0.20	kg	Hose disconnect emissions during charging

Module A4 — Transport to Installation Site

Leg	Origin	Destination	Transport Mode	Distance (km)	Mass (kg)
A4-01	Chicago IL Plant	Dallas TX Staging Yard	Heavy Lorry >32t (EURO 6)	750	4,555
A4-02	Dallas TX Staging Yard	Building Mechanical Room	Lorry 16-32t (EURO 6)	25	4,555

Transported mass includes 4,420 kg unit + 135 kg packaging. Capacity utilisation: 65%; empty return trip excluded.

Module A5 — Installation & Commissioning

Activity	Resource	Value	Unit	Notes
Crane Unloading & Placement	Diesel (50-ton hydraulic crane)	37.8	L	3.0 h @ 12.6 L/h
Electrical Hook-up & Commissioning	Grid Electricity (US-ERCOT)	350	kWh	Pump cycling & calibration
Pressure Decay Test	R134a Refrigerant Loss	0.50	kg	Gauge manifold release
Vibration Isolation Mounts	Neoprene Pads	45	kg	Installed with unit
Packaging Waste - Timber Skid	Wood to Recycling	120	kg	Chipped for biomass
Packaging Waste - Stretch Wrap	LDPE to Landfill	15	kg	Municipal landfill

Modules B1–B7 — Use Stage (25-Year RSL)

Module	Parameter	Value	Unit	Notes
B1 Use	Refrigerant Leak Rate	2.0	%/yr	0.90 kg/yr R134a, GWP100 1,430
B1 Use	Total Refrigerant Leakage over RSL	22.5	kg	0.90 kg/yr x 25 yrs
B2 Maintenance	Synthetic POE Lubricant Oil	25.0	kg/2 yrs	12 oil changes over RSL
B2 Maintenance	Refrigerant Top-Up	0.90	kg/yr	Replaces annual leakage
B2 Maintenance	Scheduled Maintenance Electricity	180	kWh/yr	Preventive servicing
B3 Repair	Condenser Tube Re-tubing (Copper)	45.0	kg/5 yrs	Incl. zinc anode replacement
B4 Replacement	Motor Bearings & Expansion Valve	38.0	kg	Replaced once at Year 15
B4 Replacement	VFD Capacitor Bank	12.0	kg	Replaced once at Year 12
B5 Refurbishment	Not scheduled within RSL	0	kg	Module declared, no impact
B6 Energy Use	Full Load Efficiency (100%)	0.580	kW/TR	AHRI 550/590 rating
B6 Energy Use	Part Load Efficiency (75% / 50% / 25%)	0.440 / 0.380 / 0.490	kW/TR	AHRI weights 42% / 45% / 12%
B6 Energy Use	Integrated Part Load Value (IPLV)	0.435	kW/TR	Weighted seasonal efficiency
B6 Energy Use	Annual Operating Hours	8,760	h/yr	Continuous duty (data centre)
B6 Energy Use	Annual Electricity Consumption	1,905,300	kWh/yr	500 RT x 0.435 kW/TR x 8,760 h
B7 Water Use	Cooling Tower Make-up Water	120.0	m3/yr	Evaporation, drift & blowdown

Modules C1–C4 — End of Life

Module	Parameter	Value	Unit	Notes
C1 Deconstruction	Dismantling Electricity	120	kWh	Recovery pump 35 + torch cutting 85
C1 Deconstruction	Refrigerant Recovered	41.4	kg	92% of 45.0 kg charge, AHRI 700 reclaim
C1 Deconstruction	Refrigerant Released	3.6	kg	Unrecoverable residual
C2 Waste Transport	Distance to Scrap Facility	100	km	Heavy Lorry >32t (EURO 6)
C3 Waste Processing	Recycling Rate	92.4	%	Metals sorted for re-melting
C3 Waste Processing	Incineration with Energy Recovery	3.1	%	Plastics & wiring insulation
C4 Disposal	Landfill Rate	4.5	%	PUF foam, oil sludge, gaskets
C4 Disposal	Landfilled Mass	197.1	kg	Inert / plastic mixture landfill

Module C3 — Material Recovery Streams

Material Stream	Mass (kg)	Recovery Route
Structural Steel & Cast Iron	2,790	Electric Arc Furnace (EAF) scrap re-melting
Copper Tubes & Motor Windings	720	Electrolytic copper smelting & refining
Aluminium Microchannel Coil	95	Secondary aluminium ingot re-smelting
Stainless Steel Piping	85	Stainless scrap re-melting
R134a Recovered Refrigerant	41.4	AHRI 700 reclamation & distillation

Module D — Benefits & Loads Beyond the System Boundary

Recovered Material	Recovery Rate (%)	Displaces	Credit (kg CO2e)
Secondary Steel Scrap	95.0	Primary blast-furnace hot-rolled steel	-1,850
Secondary Copper	96.0	Primary electrolytic copper cathode	-980
Secondary Aluminium	90.0	Primary smelted aluminium ingot	-380
Reclaimed R134a	92.0	Virgin R134a production	0
Net Avoided Burden			-3,210

Credits calculated net of secondary material input in A1 per EN 15804+A2 Annex D.